

Apache JMeter Response Assertion Testing Report Case Study 2

1. Objective

The objective of this case study was to validate server responses using various Response Assertions in Apache JMeter and verify whether the application responses satisfy the expected conditions.

2. Tool Used

- Apache JMeter 5.6.3
 - Java JDK
 - Windows Operating System
-

3. Test Configuration

Thread Groups Executed

- Response Assertion Response Code Thread Group
- Response Assertion Response Message Thread Group
- Response Assertion Response Header Thread Group
- Response Assertion Request Header Thread Group

Each Thread Group contains multiple HTTP Requests configured with different assertion validations.

4. Test Execution Procedure

Step 1

Created a Test Plan in Apache JMeter.

Step 2

Added multiple Thread Groups for assertion validation.

Step 3

Configured HTTP Requests.

Step 4

- Added Response Assertions to validate:
- Response Code
- Response Message
- Response Headers
- Request Headers

Step 5

Executed the test in GUI mode.

Step 6

Executed the test in Non-GUI mode using:

```
jmeter -n -t TestPlan1.jmx -l result1.jtl
```

Step 7

Generated HTML Dashboard Report using:

```
jmeter -g result1.jtl -o HTMLReport
```

5. Test Results

Response Code Assertion

- Purpose:
Validate whether the server returns the expected HTTP Status Code.
- Observation:
- Some requests matched the expected status code.
- Some requests failed due to assertion mismatch.

Status:

Partially Passed

Response Message Assertion

- Purpose:
Validate server response messages.
- Observation:
- Requests returning "OK" passed.
- Requests with different response messages failed.

Status:
Partially Passed

Response Header Assertion

- Purpose:
Validate response headers received from the server.
- Observation:
- Certain header validations were successful.
- Some requests failed because the expected header value was not found.

Status:
Partially Passed

Request Header Assertion

- Purpose:
Validate request header information.
- Observation:
- Several requests passed.
- Some requests failed because request headers did not match the assertion conditions.

Status:
Partially Passed

6. HTML Dashboard Analysis

Test and Report Information

Source File: result1.jtl

Dashboard Generated Successfully

HTML Report Created Successfully

Link : <file:///D:/JMeterTests/HTMLReport/CaseStudy2/index.html>

APDEX Analysis

Observed APDEX Score: Approximately 1.375

Interpretation:

- Good user experience for successful requests.
- Failed assertions reduced overall reliability.

Request Summary

PASS Requests: Majority of requests

FAIL Requests: Assertion failures observed

Observation:

- Requests reached the server successfully.
- Failures occurred due to assertion validation mismatches rather than server availability issues.

7. Performance Statistics

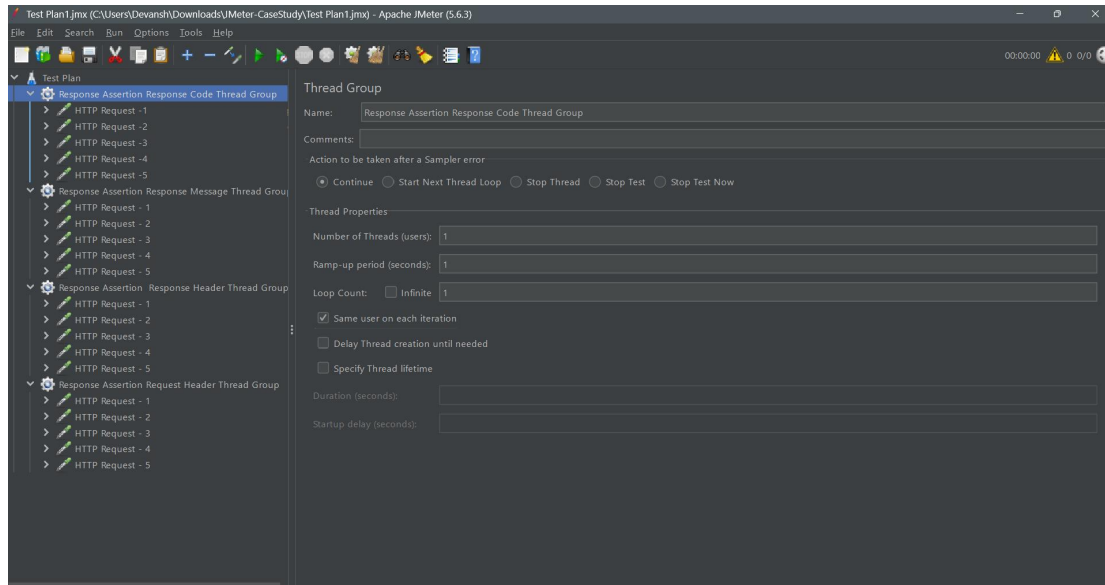
- Metrics captured from the dashboard:
- Average Response Time
- Minimum Response Time
- Maximum Response Time
- Throughput
- Error Percentage
- APDEX Score

The application remained responsive during execution.

8. Testing Link

Link : <https://practicetestautomation.com/practice-test-login/>

9. Screen Shots



Apache JMeter Dashboard

Dashboard

Let Charts

Let Customs Graphs

Test and Report information

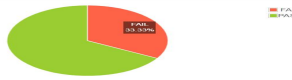
Source file
Start Time
End Time
Filter for display

result1.jfr
6/15/2026, 11:57 PM
6/15/2026, 11:57 PM
=

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.575	500 ms	1 sec 500 ms	Total
1.000	500 ms	1 sec 500 ms	HTTP Request - 5-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 5-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 5
1.000	500 ms	1 sec 500 ms	HTTP Request - 4-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 4-2
0.000	500 ms	1 sec 500 ms	HTTP Request - 3-3
0.933	500 ms	1 sec 500 ms	HTTP Request - 3-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 3-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 2-3
1.000	500 ms	1 sec 500 ms	HTTP Request - 2-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 2-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 2
0.500	500 ms	1 sec 500 ms	HTTP Request - 1-1
0.167	500 ms	1 sec 500 ms	HTTP Request - 1-2
0.000	500 ms	1 sec 500 ms	HTTP Request - 1
1.000	500 ms	1 sec 500 ms	HTTP Request - 0-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 0-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 0
1.000	500 ms	1 sec 500 ms	HTTP Request - 4-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 4-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 4
1.000	500 ms	1 sec 500 ms	HTTP Request - 3-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 3-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 3
1.000	500 ms	1 sec 500 ms	HTTP Request - 2-1
1.000	500 ms	1 sec 500 ms	HTTP Request - 2-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 2
0.500	500 ms	1 sec 500 ms	HTTP Request - 1-1
0.500	500 ms	1 sec 500 ms	HTTP Request - 1-0
0.000	500 ms	1 sec 500 ms	HTTP Request - 1

Requests Summary



Statistics

Requests	Executions				Response Times (ms)							Throughput		Network (KB/sec)	
	Label	#Execution	Fail.	Error %	Average	Min	Max	Median	95th pct	99th pct	99th pct	Transactions/s	Received	Sent	
HTTP Request - 1	1	1	0	0.00%	2220.00	2220	2220	2220.00	2220.00	2220.00	2220.00	0.45	54.28	0.13	
HTTP Request - 1-0	1	1	0	0.00%	1495.00	1495	1495	1495.00	1495.00	1495.00	1495.00	0.67	0.05	0.09	
HTTP Request - 1-1	1	1	0	0.00%	724.00	724	724	724.00	724.00	724.00	724.00	1.38	164.48	0.19	
HTTP Request - 1-2	1	1	1	100.00%	863.00	863	863	863.00	863.00	863.00	863.00	1.16	139.62	0.32	
HTTP Request - 2-0	1	1	0	0.00%	443.00	443	443	443.00	443.00	443.00	443.00	2.26	3.18	0.32	
HTTP Request - 2-1	1	1	0	0.00%	420.00	420	420	420.00	420.00	420.00	420.00	2.38	263.53	0.33	
HTTP Request - 3	1	1	1	100.00%	791.00	791	791	791.00	791.00	791.00	791.00	1.26	152.32	0.35	
HTTP Request - 3-0	1	1	0	0.00%	441.00	441	441	441.00	441.00	441.00	441.00	2.27	3.19	0.32	
HTTP Request - 3-1	1	1	0	0.00%	350.00	350	350	350.00	350.00	350.00	350.00	2.86	340.22	0.40	
HTTP Request - 4	1	1	1	100.00%	787.00	787	787	787.00	787.00	787.00	787.00	1.27	153.12	0.36	
HTTP Request - 4-0	1	1	0	0.00%	432.00	432	432	432.00	432.00	432.00	432.00	2.31	3.29	0.32	
HTTP Request - 4-1	1	1	0	0.00%	355.00	355	355	355.00	355.00	355.00	355.00	2.82	335.44	0.40	
HTTP Request - 5	1	1	1	100.00%	776.00	776	776	776.00	776.00	776.00	776.00	1.29	155.26	0.36	
HTTP Request - 5-0	1	1	0	0.00%	434.00	434	434	434.00	434.00	434.00	434.00	2.30	3.24	0.32	
HTTP Request - 5-1	1	1	0	0.00%	342.00	342	342	342.00	342.00	342.00	342.00	2.92	348.19	0.41	
HTTP Request - 1	3	3	3	100.00%	2168.67	2059	2306	2141.00	2306.00	2306.00	2306.00	1.30	156.75	0.36	
HTTP Request - 1-0	3	3	0	0.00%	1534.67	1495	1568	1541.00	1568.00	1568.00	1568.00	1.91	2.70	0.27	
HTTP Request - 1-1	3	3	0	0.00%	633.00	563	737	599.00	737.00	737.00	737.00	3.70	441.04	0.52	
HTTP Request - 1-2	3	3	3	100.00%	761.33	724	869	751.00	869.00	869.00	869.00	2.69	324.48	0.75	
HTTP Request - 2-0	3	3	0	0.00%	425.00	402	449	424.00	449.00	449.00	449.00	4.32	6.08	0.60	
HTTP Request - 2-1	3	3	0	0.00%	356.33	322	420	327.00	420.00	420.00	420.00	4.21	501.75	0.59	
HTTP Request - 3	3	3	3	100.00%	983.00	767	1367	795.00	1367.00	1367.00	1367.00	1.69	203.42	0.47	
HTTP Request - 3-0	3	3	0	0.00%	434.33	414	452	437.00	452.00	452.00	452.00	3.56	5.02	0.50	
HTTP Request - 3-1	3	3	0	0.00%	548.33	353	934	358.00	934.00	934.00	934.00	2.20	202.11	0.31	
HTTP Request - 3-2	3	3	3	100.00%	754.00	718	791	753.00	791.00	791.00	791.00	1.66	200.60	0.47	
HTTP Request - 4-0	3	3	0	0.00%	420.67	397	439	426.00	439.00	439.00	439.00	2.07	2.91	0.29	
HTTP Request - 4-1	3	3	0	0.00%	333.00	320	352	327.00	352.00	352.00	352.00	2.14	254.46	0.30	
HTTP Request - 5	3	3	3	100.00%	762.33	720	794	773.00	794.00	794.00	794.00	1.80	192.68	0.45	
HTTP Request - 5-0	3	3	0	0.00%	425.00	409	442	424.00	442.00	442.00	442.00	1.97	2.77	0.27	
HTTP Request - 5-1	3	3	0	0.00%	337.33	311	352	349.00	352.00	352.00	352.00	2.04	243.35	0.29	

Errors

Type of error
Test failed: request headers expected to contain /Connection: keep-alive; test:practicetestautomation.com\nUser-Agent: Apache-HttpClient/4.5.14 (Java/22/)
Test failed: headers expected to contain HTTP/1.1 201 OK\nDate: Mon, 15 Jun 2026 17:31:57 GMT\nContent-Type: text/html; charset=UTF-8\nTransfer-Encoding: must-revalidate\nReport-To: {\"group\":\"cf-nel\",\"max_age\":604800,\"endpoints\":[\"url\":\"https://a.nel.cloudflare.com/report/v4?e=veGZNRRLJspQvRv3C4FvYvCCKfR8Lm2FNMBC9z2BUcWzBq2MqSGCmG7VwV7YvYf5RoxvYVwzVt8qGx8zA2y8VwLTCmqd9S2FpAkyrhkFdy modified: Mon, 15 Jun 2026 17:02:10 GMT\nVhost-header: cf\nnel:cf-nel;success_fraction=0.0,max_age=604800\nServer-Timing: cfCacheHit cfDp=duration=6,cfOrigin=382nCF-4AY, cf3cf4bdebf0e6a-SIN\nvml-svc: h3=443\",ma=86400/
Test failed: headers expected to contain HTTP/1.1 201 OK\nDate: Mon, 15 Jun 2026 17:31:57 GMT\nContent-Type: text/html; charset=UTF-8\nTransfer-Encoding: Control: max-age=3, must-revalidate\nReport-To: {\"group\":\"cf-nel\",\"max_age\":604800,\"endpoints\":[\"url\":\"https://a.nel.cloudflare.com/report/v4?e=veGZNRRLJspQvRv3C4FvYvCCKfR8Lm2FNMBC9z2BUcWzBq2MqSGCmG7VwV7YvYf5RoxvYVwzVt8qGx8zA2y8VwLTCmqd9S2FpAkyrhkFdy modified: Mon, 15 Jun 2026 17:02:10 GMT\nVhost-header: cf\nnel:cf-nel;success_fraction=0.0,max_age=604800\nServer-Timing: cfCacheHit cfDp=duration=6,cfOrigin=382nCF-4AY, cf3cf4bdebf0e6a-SIN\nvml-svc: h3=443\",ma=86400/
Test failed: message expected to contain /401/
Test failed: code expected to contain 400/
Test failed: message expected to contain /401/
Test failed: headers expected to contain HTTP/1.1 201 OK\nDate: Mon, 15 Jun 2026 17:31:57 GMT\nContent-Type: text/html; charset=UTF-8\nTransfer-Encoding: Encoding: Cookie\nCache-Control: max-age=3, must-revalidate\nReport-To: {\"group\":\"cf-nel\",\"max_age\":604800,\"endpoints\":[\"url\":\"https://a.nel.cloudflare.com/report/v4?e=veGZNRRLJspQvRv3C4FvYvCCKfR8Lm2FNMBC9z2BUcWzBq2MqSGCmG7VwV7YvYf5RoxvYVwzVt8qGx8zA2y8VwLTCmqd9S2FpAkyrhkFdy modified: Mon, 15 Jun 2026 17:02:10 GMT\nVhost-header: cf\nnel:cf-nel;success_fraction=0.0,max_age=604800\nServer-Timing: cfCacheHit cfDp=duration=6,cfOrigin=382nCF-4AY, cf3cf4bdebf0e6a-SIN\nvml-svc: h3=443\",ma=86400/
Test failed: message expected to contain /401/
Test failed: request headers expected to contain /Connection: keep-alive; test:practicetestautomation.com\nUser-Agent: Apache-HttpClient/4.5.14 (Java/22/)
Test failed: headers expected to contain HTTP/1.1 201 OK\nDate: Mon, 15 Jun 2026 17:31:57 GMT\nContent-Type: text/html; charset=UTF-8\nTransfer-Encoding: Encoding: Cookie\nCache-Control: max-age=3, must-revalidate\nReport-To: {\"group\":\"cf-nel\",\"max_age\":604800,\"endpoints\":[\"url\":\"https://a.nel.cloudflare.com/report/v4?e=veGZNRRLJspQvRv3C4FvYvCCKfR8Lm2FNMBC9z2BUcWzBq2MqSGCmG7VwV7YvYf5RoxvYVwzVt8qGx8zA2y8VwLTCmqd9S2FpAkyrhkFdy modified: Mon, 15 Jun 2026 17:02:10 GMT\nVhost-header: cf\nnel:cf-nel;success_fraction=0.0,max_age=604800\nServer-Timing: cfCacheHit cfDp=duration=6,cfOrigin=382nCF-4AY, cf3cf4bdebf0e6a-SIN\nvml-svc: h3=443\",ma=86400/
Test failed: message expected to contain /401/
Test failed: request headers expected to contain /Connection: keep-alive; test:practicetestautomation.com\nUser-Agent: Apache-HttpClient/4.5.14 (Java/22/)
Test failed: message expected to contain /401/
Test failed: code expected to contain 200/
Test failed: code expected to contain 200/
Test failed: request headers expected to contain /Connection: keep-alive; test:practicetestautomation.com\nUser-Agent: Apache-HttpClient/4.5.14 (Java/22/)
Test failed: headers expected to contain HTTP/1.1 201 OK\nDate: Mon, 15 Jun 2026 17:31:57 GMT\nContent-Type: text/html; charset=UTF-8\nTransfer-Encoding: Encoding: Cookie\nCache-Control: max-age=3, must-revalidate\nReport-To: {\"group\":\"cf-nel\",\"max_age\":604800,\"endpoints\":[\"url\":\"https://a.nel.cloudflare.com/report/v4?e=veGZNRRLJspQvRv3C4FvYvCCKfR8Lm2FNMBC9z2BUcWzBq2MqSGCmG7VwV7YvYf5RoxvYVwzVt8qGx8zA2y8VwLTCmqd9S2FpAkyrhkFdy modified: Mon, 15 Jun 2026 17:02:10 GMT\nVhost-header: cf\nnel:cf-nel;success_fraction=0.0,max_age=604800\nServer-Timing: cfCacheHit cfDp=duration=6,cfOrigin=382nCF-4AY, cf3cf4bdebf0e6a-SIN\nvml-svc: h3=443\",ma=86400/
Test failed: message expected to contain /401/
Test failed: request headers expected to contain /Connection: keep-alive; test:practicetestautomation.com\nUser-Agent: Apache-HttpClient/4.5.14 (Java/22/)
Test failed: message expected to contain /401/
Test failed: request headers expected to contain /Connection: keep-alive; test:practicetestautomation.com\nUser-Agent: Apache-HttpClient/4.5.14 (Java/22/)

10. Error Analysis

The HTML Dashboard Error Section indicates:

- Assertion comparison failures.
- Expected values did not match actual server responses.
- Some Response Code, Response Message, Header validations failed.

Root Cause:

The assertion conditions configured in JMeter did not exactly match the actual response returned by the application.

Examples:

- Incorrect expected Response Code.
- Incorrect expected Response Message.
- Missing Response Header.
- Request Header mismatch.

11. Observations

1. Test Plan executed successfully.
2. JMeter generated result1.jtl successfully.
3. HTML Dashboard report was generated successfully.
4. Application server responded to all requests.
5. Failures were primarily due to assertion validation mismatches.
6. No JMeter execution errors were observed.
7. Assertion configuration requires correction for failed requests.

12. Conclusion

The Response Assertion Case Study was executed successfully using Apache JMeter 5.6.3.

The test validated Response Codes, Response Messages, Response Headers, and Request Headers. While the application responded successfully, several assertion validations failed because the expected values configured in JMeter did not match the actual values returned by the server.

The case study successfully demonstrates the implementation and validation of Response Assertions in Apache JMeter and highlights the importance of configuring accurate assertion criteria for reliable test results.

Result: Executed Successfully

Overall Status: Assertion Validation Completed with Mixed Pass and Fail Results